• Apply the data analysis and visualization tools / techniques to analyze and visualize the pattern in the data.	**5. Data Analysis (12 hr)** 5.1. Quantitative Data Analysis 5.1.1. Types of Quantitative Data 5.1.2. Data Coding 5.1.3. Data Presentation: Tables, Bar charts, Scatter Graph, Line graph 5.1.4. Analysis Techniques: Descriptive Statistics, Inferential Statistics (Z-test, t-test, Chi-square test, ANOVA), Regression Analysis, Time Series Analysis and Multivariate Analysis 5.2. Qualitative Data Analysis 5.2.1. Qualitative Data 5.2.2. Analyzing Textual Data 5.2.3. Analyzing Non-Textual Data 5.3. A detailed Case Study in data visualization and analysis using Python libraries: Pandas, NumPy, Matplotlib, Seaborn, SciPy, Scikit-learn

1. What is time series data? Explain the basic techniques used in time series analysis.
2. How is the t-test different from the z-test? Explain ANOVA and its significance in research.

The differences between the **t-test** and the **z-test** primarily relate to sample size and knowledge of population parameters, while **ANOVA** serves as a vital tool for comparing multiple groups simultaneously.

Differences Between the t-Test and z-Test

Both tests are used to determine if there is a significant difference between means, but they are applied under different conditions:

Feature	z-Test	t-Test
Sample Size	Used for large samples where **n ≥ 30**.	Used for small samples where **n < 30**.
Population Variance (σ)	Must be **known** and not derived from the data.	Used when the population variance is **unknown**.

Distribution	Relies on the **Standard Normal Distribution**.	Relies on the **t-distribution**, which varies based on **degrees of freedom**.
Standard Error Calculation	Uses the population standard deviation: $\sigma_{\bar{x}} = \sigma / \sqrt{n}$.	Uses the sample standard deviation (**s**) as an estimate: $\sigma_{\bar{x}} = s / \sqrt{n}$.

Key Decision Rule: If the sample size is at least 30, the sampling distribution is considered normal, and the z-test is appropriate. If the sample is small and the population is normal, the t-test is used.

ANOVA (Analysis of Variance)

Definition: ANOVA is a statistical method used to test for differences between the means of **two or more independent groups**. It is considered an **extension of the t-test**; in fact, an ANOVA performed on only two groups is mathematically equivalent to a t-test.

The F-Statistic: ANOVA calculates an **F-statistic**, which is a ratio comparing the variability **between groups** (due to the treatment) against the variability **within groups** (the "background noise" or error).

- **H_0 (Null Hypothesis):** All population means are equal ($\mu_1 = \mu_2 = \mu_3 ...$).
- **H_1 (Alternative Hypothesis):** At least one of the population means is different.

Significance of ANOVA in Research

1. **Control of Type I Error:** If a researcher has three or more groups, performing multiple pairwise t-tests increases the probability of a **Type I error** (falsely claiming a difference exists). ANOVA allows for testing these differences "all at once" without inflating the error rate.
2. **Partitioning of Variance:** ANOVA provides a "blueprint" for research by breaking down total variation (**TSS**) into components: variation between groups (**SSB**) and variation within groups (**SSW**). This helps researchers isolate the specific impact of a treatment.
3. **Determining Explanatory Power:** Through the **Coefficient of Determination (R^2)**, ANOVA quantifies how much of the variance in the outcome (dependent variable) is actually explained by the independent variable or treatment group.
4. **Initial Screening for Differences:** A significant ANOVA result acts as a trigger for further investigation. While it only confirms that *at least* two groups differ, it signals the

need for more sophisticated "post-hoc" analyses to identify exactly which groups are different.
5. **Efficiency in Complex Designs:** Beyond one-way analysis, ANOVA can accommodate multiple independent factors or treatments simultaneously, provided they are independent.

3. Describe the role of multivariate analysis in understanding complex IT data, including server performance indicators, user activity, and transaction records. Further, discuss how techniques such as Principal Component Analysis (PCA) help reduce dimensionality and make the results easier to interpret.
4. Explain the Chi-Square test with an example.

The **Chi-Square** (χ^2) **test** is an inferential statistical technique used to determine if there is a significant association between **categorical variables**. Unlike tests that compare means (like the t-test), the Chi-Square test focuses on frequencies and proportions within specific categories.

Types of Chi-Square Tests

According to the sources, there are two primary types of Chi-Square tests:

- **Test of Independence:** This assesses whether observed frequencies are contingent upon certain conditions, determining if there is a significant relationship between two categorical variables. An example would be testing if gender (male/female) is related to product preference (like/dislike).
- **Goodness-of-Fit Test:** This compares observed frequencies in a data sample against frequencies based on a prior expectation (either theoretical or empirical). For instance, a researcher might test if the distribution of blood types in a specific sample matches the known distribution in the general population.

Mathematical Procedure

The test follows a systematic process to evaluate the **null hypothesis**, which assumes that the variables are independent (i.e., there is no relationship).

1. **Calculate Expected Frequencies:** Under the assumption of independence, expected counts for each category are calculated by multiplying marginal probabilities by the total sample size (N).

2. **Apply the Formula:** The Chi-Square statistic is calculated by summing the squared difference between observed and expected values, divided by the expected value for

$$\chi^2 = \sum \frac{(observed - expected)^2}{expected}$$

each cell:
3. **Determine Degrees of Freedom (\(df\)):** This is calculated based on the table structure

$$df = (rows - 1) \times (columns - 1)$$

using the formula:
4. **Evaluate Significance:** A rule of thumb is that if the calculated Chi-Square statistic is much greater than the degrees of freedom, it indicates **statistical significance**.

Example: Social Conformity and Group Size

The sources provide a detailed real-world example based on a visual perception study regarding social pressure and conformity.

5. **The Problem:** Researchers wanted to know if the size of a group (number of "conspirators" giving wrong answers) alters the proportion of subjects who conform to an obviously incorrect group consensus.
6. **The Data:** In 500 experiments across group sizes of 2, 4, 6, 8, and 10 members, the observed number of subjects who conformed were:
    a. **Group Size 2:** 20 conformed
    b. **Group Size 4:** 50 conformed
    c. **Group Size 6:** 75 conformed
    d. **Group Size 8:** 60 conformed
    e. **Group Size 10:** 30 conformed
7. **Null Hypothesis (\(H_0\)):** Group size and conformity are independent; group size has no effect on the likelihood of a subject conforming.
8. **Expected Frequencies:** The overall likelihood of conforming was calculated as 235/500 = **0.47**. If group size had no effect, researchers would expect exactly 47 subjects to conform in every group size category.
9. **Test Results:** Using the formula to compare these observed values against the expected value of 47, the researchers found a **Chi-Square statistic of approximately 85**.
10. **Conclusion:** With \(df = 4\), the statistic (85) was much higher than the degrees of freedom. Therefore, the researchers rejected the null hypothesis and concluded that group size **does** significantly alter the proportion of subjects who conform.

11. Critically evaluate the role of inferential and descriptive statistics in quantitative research. How do these methods contribute to hypothesis testing and decision-making?

In quantitative research, statistics serve as the primary tool for converting raw data into meaningful information used for generalization and decision-making. The methodology is divided into two major roles: **descriptive statistics**, which organize and summarize data, and **inferential statistics**, which allow researchers to draw conclusions about a population based on sample findings.

1. The Role of Descriptive Statistics

Descriptive statistics consist of methods for **organizing, exploring, and summarizing** information. Their primary goal is to provide a clear "snapshot" or portrayal of the data characteristics without making claims beyond the immediate sample.

- **Organization and Visualization:** This includes the construction of frequency distributions, tables, and graphs such as pie charts, bar graphs, and line graphs.
- **Measures of Center:** These indicate the "typical" value in a dataset, including the **mean** (average), **median** (middle value), and **mode** (most frequent value).
- **Measures of Variation:** These describe the spread of the data, including the **range**, **standard deviation**, and **inter-quartile range (IQR)**.
- **Application in Computer Science (Phishing):** A researcher might use descriptive statistics to report that 45% of URLs in a specific dataset were identified as phishing attempts, or to show a bar graph comparing the lengths of legitimate versus malicious URLs.

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2. The Role of Inferential Statistics

Inferential statistics go beyond description by providing methods for **generalizing** from a sample to a population with a calculated degree of certainty. Because it is often impossible to study every individual in a population (e.g., every phishing website ever created), researchers use sample statistics as estimates for unknown population parameters.

- **Estimation:** This involves using sample data to form **point estimates** (a single best guess for a parameter) or **interval estimates** (a range of numbers, such as a **Confidence Interval**, within which the parameter is believed to fall).
- **Measuring Reliability:** Inferential statistics provide the mathematical framework to measure the reliability of conclusions, helping researchers assess the strength of their findings and evaluate uncertainty.
- **Application in Computer Science (Phishing):** If a researcher finds that a new detection tool is 99% accurate on a sample of 1,000 URLs, they use inferential statistics

to predict how reliably that tool will perform against the millions of phishing sites in the "real world" population.

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3. Contribution to Hypothesis Testing

Hypothesis testing is a systematic procedure for testing claims about population parameters using sample statistics. In quantitative research, this often follows a **deductive approach**, where a researcher starts with a theory and specifies a narrow hypothesis to test.

- **Null (H_0) vs. Alternative (H_a) Hypotheses:** The researcher assumes the null hypothesis is true (e.g., "There is no difference between detection methods") and seeks evidence to reject it in favor of the alternative hypothesis.
- **Statistical Tests:** Depending on the data type (categorical or continuous) and sample size, researchers use different tests:
    - **Z-test:** For large samples (**$n \geq 30$**) with a known population standard deviation.
    - **t-test:** For smaller samples (**$n < 30$**) to compare the means of two groups.
    - **ANOVA:** Used to compare the means of three or more independent groups simultaneously.
    - **Chi-square Test:** Used to determine if there is a significant association between categorical variables.
- **Example:** A researcher might hypothesize that "A new detection algorithm enables faster data processing than the current algorithm". They would then use a t-test to determine if the measured difference in speed is statistically significant.

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4. Contribution to Decision-Making

Statistics provide a formal approach to **decision-making under uncertainty**. Researchers use criteria to decide whether to retain or reject their claims, thereby minimizing the risk of errors.

- **Level of Significance (α):** This is the maximum allowable probability of making a **Type I error** (rejecting a true null hypothesis). Common levels are 0.05 or 0.01.
- **P-value Method:** The P-value is the probability of obtaining the observed results if the null hypothesis were true. If the **P-value ≤ α**, the researcher rejects the null hypothesis and concludes the findings are statistically significant.
- **Error Management:** Statistics help distinguish between a "Correct Decision" and errors. A **Type II error** occurs when a researcher fails to reject a false null hypothesis.
- **Phishing Detection Context:** Based on the results of a Z-test, a security firm might **decide** to deploy a new browser plugin because the statistical evidence strongly supports its superior accuracy compared to existing blacklists.

12. Describe Grounded Theory, Distributed Cognition, and Activity Theory, and explain their significance in qualitative analysis.

Based on the provided sources, **Grounded Theory** is a well-documented qualitative strategy. However, descriptions of **Distributed Cognition** and **Activity Theory** are not contained within the provided excerpts. Descriptions for those two theories are provided using information from outside the sources for completeness.

Grounded Theory

As described in the sources, Grounded Theory is a qualitative strategy of inquiry where the researcher derives a **general, abstract theory** of a process, action, or interaction that is "grounded" in the views of the participants.

- **Characteristics:** This design is distinguished by two primary features: the **constant comparison** of data with emerging categories and **theoretical sampling** of different groups to maximize the similarities and differences of information.
- **The Process:** Unlike quantitative research that begins with a theory to test, Grounded Theory uses an **inductive process** where theory is the end point of the study. The researcher collects data in multiple stages, refining and interrelating categories of information as they surface.
- **Analysis Steps:** Grounded Theory involves a systematic coding process:
    1. **Open Coding:** Generating initial categories of information.
    2. **Axial Coding:** Selecting one category and positioning it within a theoretical model.
    3. **Selective Coding:** Explicating a story from the interconnection of these categories.
- **Significance:** It is significant in qualitative analysis because it moves beyond mere description to build a **theoretical model** or logic diagram that explains a central phenomenon. It is particularly useful for studying topics where existing theories do not apply or where little research has been conducted.

Distributed Cognition and Activity Theory

(Note: The following information is not found in the provided sources and should be independently verified.)

While these frameworks are not detailed in your sources, they are frequently used in Computer Science and Human-Computer Interaction (HCI) research—fields mentioned in your syllabus.

- **Distributed Cognition:** This theory argues that cognition is not restricted to an individual's mind but is distributed across people, artifacts (tools, software), and environments.

    - **Significance:** In qualitative analysis, it is significant for researchers studying collaborative work environments (like an IT help desk or a software development team). It helps analyze how information is transformed and propagated through a system of people and tools.
- **Activity Theory:** This is a conceptual framework that treats an "activity" as the primary unit of analysis. It examines the relationship between a subject (the human), an object (the goal), and the tools, rules, and community that mediate the work.
    - **Significance:** It is significant for understanding the complex social and technical context of system use. Qualitative researchers use it to identify "contradictions" within a system (e.g., a new security policy that conflicts with a user's need for speed), which often drive the evolution of technology.

Significance in Qualitative Analysis

Collectively, these theories provide the "Paradigm" or **shared way of thinking** required for high-quality qualitative research. They are significant because they:

- **Provide a Lens:** They act as a theoretical lens that shapes the types of questions asked and how data is interpreted.
- **Support Inductive Logic:** They offer structured methods to build from specific observations to broad patterns and generalizations.
- **Enhance Credibility:** By following the rigorous procedural steps of these theories (like Grounded Theory's coding stages), researchers can demonstrate that their findings are systematic and reliable.

13. Why is regression analysis important in research? Briefly explain simple linear and multivariate regression.

Regression analysis is a fundamental quantitative technique used in research to explore the nature of relationships between variables, identify functional dependencies, and make reliable predictions about future outcomes.

Importance of Regression Analysis in Research

Regression analysis is essential for several reasons:

- **Predictive Modeling:** It establishes a mathematical relationship (function) that allows researchers to predict the values of a dependent variable (Y) based on given values of an independent variable (X).
- **Explaining Variability:** Through measures like the **Coefficient of Determination (R^2)**, researchers can quantify how much of the variation in an outcome is explained by specific predictors.

- **Testing Relationships:** It provides a formal method to determine if a statistically significant relationship actually exists between variables, rather than just an observed association.
- **Controlling for Biases:** It helps account for the **"regression to the mean"** effect—the tendency for extreme values to move toward the average in subsequent measurements—which can otherwise lead to misleading results if not recognized.

Simple Linear Regression

Simple linear regression is the most basic form of regression analysis, involving **one independent variable** and **one dependent variable**. It assumes that the relationship between the variables can be represented by a straight line.

- **Mathematical Form:** It is typically expressed as $Y = b_0 + b_1X + \epsilon$, where b_0 is the intercept, b_1 is the slope, and ε is an error term representing measurement or specification errors.
- **Goal:** The primary objective is to use the **Method of Least Squares** to find the line that minimizes the difference between observed data points and their predictions.

Multivariate (Multiple) Regression

Multivariate regression (often referred to in the sources as **Multiple Regression**) involves **two or more independent variables** used to predict a single dependent variable.

- **Applications:** It is used when a researcher wants to see the joint impact of several factors on an outcome. For example, a study might regress "intent to leave a college" against organizational, personal, and environmental variables simultaneously.
- **Key Techniques:**
    - **Multiple Linear Regression:** Used when the dependent variable is continuous and normally distributed.
    - **Logistic Regression:** A specific multivariate technique used when the outcome variable is **binary** (e.g., "yes/no" or "success/failure"), providing adjusted odds ratios for the predictors.
- **Significance:** This approach is vital for identifying which specific factors among many are most strongly associated with the research outcome.

14. How do you analyze textual and non-textual data? Explain briefly.

The analysis of data depends on whether the information is **textual** (words and stories) or **non-textual** (images, sounds, and numbers). In research, these methods are often divided into **qualitative** and **quantitative** approaches to identify patterns and draw meaningful conclusions.

1. Analyzing Textual Data

Analyzing textual data primarily involves making sense of written or spoken words to identify themes and meanings.

- **Transcription:** Before analysis, audio recordings (like interviews) are converted into **written form**. This allows the researcher to search and organize the data effectively.
- **Coding:** This is the process of organizing the material into "chunks" or segments of text. Researchers label these segments with terms—often the actual language used by the participant (called **in vivo terms**).
- **Thematic Analysis:** A qualitative approach used to explore and identify **recurring themes** or patterns across multiple documents or transcripts.
- **Content Analysis:** A quantitative way of analyzing text by **counting the occurrences** of specific elements, such as how many times a specific security warning appears in a set of technical reports.
- **Generic Process:** The researcher typically follows a cycle: **Organize/Prepare data** \(\rightarrow\) **Read through all data** \(\rightarrow\) **Code the data** \(\rightarrow\) **Generate themes** \(\rightarrow\) **Interpret meaning**.

2. Analyzing Non-Textual Data

Non-textual data includes **multimedia documents** such as images, photographs, diagrams, sounds (aural), and electronic sources like screenshots.

- **Semiotic Analysis:** This technique is specifically used for **images** to study the symbolism and embedded meanings within the visual content.
- **Visual Content Analysis:** Treating documents as **objects**, researchers examine how they are produced, consumed, and circulated.
- **Data Transformation (Quantifying):** In mixed-methods research, non-textual or qualitative data is often **quantified** by counting the frequency of specific visual elements so they can be compared using statistical software.
- **Aural Analysis:** This involves analyzing **sounds or music** to capture data that might be missed by observation alone.

Simple Comparison Table

Feature	Textual Analysis	Non-Textual Analysis
Primary Sources	Interviews, diaries, reports.	Photos, art, sounds, videos.

Key Technique	**Coding** and **Thematic Analysis**.	**Semiotic Analysis** and **Visual Mapping**.
Goal	To understand meanings and experiences.	To interpret symbolism and visual patterns.

Example in Phishing Website Detection

- **Textual Analysis:** A researcher analyzes the **textual content** of phishing emails to identify common psychological "lures" (e.g., urgency, fear) using **thematic analysis**.
- **Non-Textual Analysis:** A researcher uses **semiotic analysis** to study the **visual logos** and color schemes used on fraudulent websites to see how they mimic legitimate brands.

15. Differentiate between textual and non-textual qualitative data. Discuss the role of coding and thematic analysis in extracting meaningful insights from IT user feedback or software bug reports.

Qualitative research relies on diverse forms of data to explore and understand the meanings individuals ascribe to problems. In Information Technology (IT) research, this often involves analyzing user feedback or software bug reports.

Differentiation Between Textual and Non-Textual Qualitative Data

The primary distinction between these data types lies in their format and the senses they engage:

- Textual Qualitative Data: This consists of written or spoken words. In an IT context, common sources include:
    - User Feedback: Comments provided in surveys, emails, or online forums.
    - Software Bug Reports: Detailed written accounts of technical issues, system behavior, and user interactions.
    - Documentation: Technical manuals, project reports, and meeting minutes.
- Non-Textual Qualitative Data: This includes "multimedia documents" such as images, sounds, and electronic captures. In IT, these often supplement textual data to provide a "holistic account" of an issue. Examples include:
    - Visual Sources: Screenshots of software errors, user interface diagrams, and videos of users interacting with a system.

    - Aural Sources: Recordings of user "think-aloud" sessions during usability testing.
    - Electronic Sources: Online community archives and screenshots of specific website types.

Role of Coding in Extracting Insights

Coding is the critical initial step in qualitative data analysis, defined as the process of organizing raw material into "chunks" or segments before bringing meaning to the information.

1. Organization: Researchers take text data (like a bug report) and segment sentences or paragraphs into categories. For example, a coder might label segments as "UI glitch," "data loss," or "user frustration."
2. Labeling (In Vivo Coding): Labels are often based on the actual language used by the IT user, known as in vivo terms. This ensures the analysis remains grounded in the user's perspective rather than the researcher's bias.
3. Codebooks: In systematic IT research, investigators may use a qualitative codebook—a table containing predetermined codes, their definitions, and specific instances (like line numbers in a bug report) where they appear. This is especially valuable when multiple team members are analyzing a large volume of feedback.

Role of Thematic Analysis in Extracting Insights

Thematic Analysis is the process of identifying and exploring recurring themes or patterns that cut across multiple data sources, such as a collection of feedback from different users.

1. Inductive Building: Themes are built from the "bottom up" by organizing codes into increasingly abstract units of information. For instance, multiple codes related to "slow login," "lagging menus," and "timed-out sessions" might be grouped under a single theme: "System Latency".
2. Identifying Major Findings: Researchers typically aim to identify five to seven major themes that represent the core findings of the study. These themes often become the primary headings in the research report.
3. Complex Theme Connections: Sophisticated analysis goes beyond identifying themes to interconnect them into a story line or a theoretical model. In software research, this might involve showing how the theme of "Inadequate Training" leads to the theme of "High User Error Rates".

4. Verification and Triangulation: Thematic analysis allows for triangulation—comparing themes derived from textual bug reports with non-textual evidence like error screenshots to confirm the accuracy of findings

16. Briefly explain the techniques used for data presentation and analysis.

Techniques for data presentation and analysis are essential for converting raw data into meaningful information and credible knowledge. These techniques are generally categorized into **quantitative**, **qualitative**, and **mixed-methods** approaches.

1. Quantitative Data Presentation and Analysis

Quantitative techniques focus on numeric data and statistical procedures to test theories or describe trends.

Data Presentation Techniques:

- **Tables:** Used to summarize data in a structured, grid-like format for easy reference.
- **Graphs and Charts:** Visual tools used to display patterns, including:
    - **Bar Charts:** Effective for comparing discrete categories or qualitative variables.
    - **Pie Charts:** Used to show the proportion of different categories in a whole dataset.
    - **Line Graphs:** Ideal for showing trends or changes in data over time.
    - **Scatter Graphs:** Used to visualize relationships between two variables.
    - **Histograms:** Used to show the frequency distribution of continuous quantitative variables.
- **Infographics:** Simplifies complex research data into visually appealing, digestible formats for both experts and non-experts.

Data Analysis Techniques:

- **Descriptive Statistics:** Methods for organizing and summarizing data characteristics, such as **measures of center** (mean, median, mode) and **measures of variation** (range, standard deviation, and inter-quartile range).
- **Inferential Statistics:** Techniques used to draw conclusions or make predictions about a population based on a sample, including **hypothesis testing** (e.g., Z-test, t-test, Chi-square test, and ANOVA).
- **Regression Analysis:** The process of fitting a mathematical line to data to explain relationships and predict future outcomes.
- **Time Series Analysis:** Identifies and visualizes patterns in data collected over an extended period.
- **Multivariate Analysis:** Simultaneously analyzes multiple variables to identify complex patterns and relationships.

2. Qualitative Data Analysis and Presentation

Qualitative techniques focus on understanding meanings, experiences, and social phenomena through non-numeric data.

Analysis Process:

- **Data Preparation:** Involves transcribing interviews, scanning documents, or typing up field notes.
- **Coding:** The process of segmenting text or images into "chunks" and labeling them with terms, often using the participant's actual language (**in vivo terms**).
- **Thematic Analysis:** Identifying and exploring recurring patterns or themes that appear across multiple data sources.
- **Content Analysis:** A quantitative approach to qualitative data that involves counting the occurrences of specific elements (e.g., specific words in a phishing email).

Representation Techniques:

- **Narrative Discussion:** Conveying findings through detailed passages, chronologies, or interconnected themes.
- **Visual Models:** Using process models, site drawings, or participant tables to supplement the text.

3. Mixed Methods Analysis Techniques

Mixed methods research integrates both quantitative and qualitative data to provide a comprehensive understanding of a problem.

- **Data Transformation:** Quantifying qualitative data (e.g., counting the frequency of themes) or qualifying quantitative data (e.g., creating themes from statistical factors) to allow for direct comparison.
- **Matrix Creation:** Combining qualitative themes and quantitative categorical variables into a grid for side-by-side comparison.
- **Outlier Exploration:** Using qualitative interviews to investigate "extreme" cases identified during initial quantitative analysis.

4. Final Presentation Forms

The results of the analysis are disseminated to the academic community through various channels:

- **Written Presentation:** Academic papers, theses, dissertations, and technical reports.
- **Visual Presentation:** Conference posters and software demonstrations.
- **Oral Presentation:** Conference talks and formal thesis defenses (**Viva**).

17. What is the significance of data visualization in quantitative analysis, and how do Matplotlib and Scikit-learn contribute to effective data presentation and interpretation decision-making in research.

In quantitative analysis, **data visualization** is significant because it allows researchers to **organize, summarize, and explore** information to identify and visualize **patterns** within a dataset. Because **90% of information transmitted to the brain is visual**, visual representations like charts and graphs capture attention more effectively than text-heavy content and increase audience retention. Presenters who use visual aids are found to be **43% more effective** in persuading audience members to take a desired course of action. Furthermore, effective visualization through tools like **infographics** can simplify complex research data, summarize lengthy datasets into concise insights, and make findings accessible to non-expert audiences.

Matplotlib and Scikit-learn contribute to this process as specialized Python libraries used in detailed **case studies** for data visualization and analysis. **Matplotlib** is primarily utilized to apply visualization techniques, enabling researchers to create **tables, bar charts, scatter graphs, and line graphs** that provide a clear "snapshot" of data characteristics. **Scikit-learn** supports the analytical component by facilitating the application of statistical methods such as **Descriptive Statistics, Inferential Statistics (e.g., Z-tests, t-tests, ANOVA), and Regression Analysis**. Together, these libraries help researchers process **technical metrics** and transform raw data into a formal record of procedures and results.

These tools enhance **decision-making** in research by providing a systematic approach to handling **uncertainty** through statistical testing. The visual outputs generated by Matplotlib allow statistical results—such as those from **hypothesis testing**—to be organized around figures that can stand on their own without the need for the entire paper to be read. For instance, a **scatterplot** can suggest a **functional relationship** between variables, which a researcher then tests using Scikit-learn to determine if the relationship is statistically significant. By visualizing **rejection regions** and **P-values**, researchers can more easily interpret whether to reject or fail to reject a **null hypothesis**. Ultimately, these libraries serve the "Presentation" phase of research by professionally conveying findings through **dissertations, conference articles, or computer-based products**, which are essential for academic scrutiny and evaluation.

18. Importance of data visualization
19. Discuss the various types of quantitative data and explain the basic techniques for data presentation.

Quantitative data consists of **numerical variables** that provide information through specific measurements or counts. In research, these variables are fundamental because they are

transformed into numerical measures for statistical treatment. Quantitative data is primarily classified into two types:

- **Discrete Variables:** These are variables whose possible values are **ordinary counting numbers** (such as 0, 1, 2, 3...). If a discrete variable has only a few different values, it can be summarized in the same way as qualitative data. Examples include **behavioral counts** (the frequency of specific actions) or **technical metrics** like the number of bugs in a piece of code.
- **Continuous Variables:** These represent quantities that can be **measured arbitrarily accurately**. Common examples include length, weight, temperature, and time-related metrics such as the time required to process data files.

Basic Techniques for Data Presentation

Data presentation involves the use of **Descriptive Statistics**, which consist of methods for organizing and summarizing collected information. The results are typically displayed in **formal records** such as technical reports or research papers to make the findings accurate, verifiable, and objective. The basic techniques include:

1. Tables

Frequency distributions are used to summarize quantitative data by listing classes (categories) alongside their frequencies or relative frequencies. Tables are highly effective for providing a structured, **grid-like summary** of observations and measurements, such as project data (person-hours and costs).

2. Graphical Representations

Visual aids are significant in research because **90% of information transmitted to the brain is visual**, making these tools more effective for persuasion and retention than text alone.

- **Bar Charts:** These are used to compare the frequencies or percentages of different categories or discrete values.
- **Pie Charts:** These circular graphs show the **proportion of each category** relative to the whole dataset.
- **Histograms:** Specifically used for quantitative variables, histograms display frequency distributions by grouping data into **class intervals** of equal length.
- **Line Graphs:** These are ideal for showing **trends and patterns over time**. They are frequently used in single-subject research designs to connect plotted data points from baseline and treatment phases.
- **Scatter Graphs (Scatter Plots):** These visualize the **association or functional relationship** between two quantitative variables, such as the relationship between a tractor's age and its maintenance cost.

- **Boxplots:** Based on the **"five-number summary"** (Minimum, Q1, Median, Q3, and Maximum), boxplots provide a graphical display of both the center and the variation of measurements.

3. Infographics

Infographics are used to simplify complex research data into **visually appealing and digestible formats**. They are particularly useful for narrating the research process and highlighting key findings for both expert and non-expert audiences by using icons and minimal text.

20. How do you choose an appropriate measure of central tendency? Explain the role of standard deviation, correlation coefficient and ANOVA in data analysis.

Choosing the appropriate measure of central tendency—**mean**, **median**, or **mode**—depends primarily on the type of data (qualitative vs. quantitative) and the distribution of that data (symmetric vs. skewed).

Choosing a Measure of Central Tendency

Measures of center indicate where the "typical" value of a variable lies within a dataset. The selection criteria are as follows:

- **Mode:** This should be used for **qualitative (categorical) variables**. It represents the value that occurs with the greatest frequency. For example, in a study of blood types, the mode would identify the most common blood type in the sample.
- **Mean:** This is the most common measure for **quantitative variables** when the distribution is **symmetric**. However, the mean is highly sensitive to **outliers** (observations that fall far from the rest of the data), which can pull the mean away from the true center.
- **Median:** This is the best choice for **quantitative variables** with a **skewed distribution**. Because the median is the middle value in an ordered list, it is not influenced by extreme outliers the way the mean is.

The Role of Standard Deviation

The **Standard Deviation (SD)** is a measure of variation that describes the spread of data around the mean. Its roles include:

- **Measuring Dispersion:** It provides a kind of average of how much observed values deviate from the mean. This helps researchers distinguish between two datasets that might have the same average but different levels of variability.

- **Applying the Empirical Rule:** For bell-shaped, symmetric distributions, the SD determines the proportion of data within specific ranges: approximately **68%** of values fall within 1 SD of the mean, **95%** within 2 SDs, and **99.7%** within 3 SDs.
- **Calculating Standard Error:** It is a key component in determining the **standard error** of a sample mean $(\sigma/\sqrt{n})$, which measures how much the sample statistic is expected to vary from the true population parameter.

The Role of the Correlation Coefficient

The correlation coefficient is a quantitative technique used to explore relationships between variables. Its roles include:

- **Identifying Linear Relationships:** Specifically, the **Pearson product-moment correlation** is used to show the strength and direction of the linear association between two continuous, normally distributed variables.
- **Checking Assumptions:** In regression analysis, correlations are used to verify if the data meets the necessary assumptions for the model to be valid.
- **Non-Parametric Analysis:** If the data is not normally distributed, researchers use the **Spearman rank-order correlation** as an alternative to assess associations.

The Role of ANOVA (Analysis of Variance)

ANOVA is a statistical method used to test for significant differences between the means of **three or more independent groups**. Its roles in analysis are:

- **Controlling Type I Error:** If a researcher compared three groups using multiple pairwise t-tests, the chance of making a Type I error (falsely claiming a difference) would increase. ANOVA allows for comparing all groups "all at once" to maintain a stable significance level.
- **Partitioning of Variance:** ANOVA breaks total variability into two parts: **variability between groups** (caused by the treatment or independent variable) and **variability within groups** (background noise or chance error).
- **Determining Explanatory Power:** Through the **Coefficient of Determination (\(R^2\))**, ANOVA tells the researcher how much of the variation in the dependent variable is actually explained by the independent variable.
- **Significance Screening:** A statistically significant ANOVA result indicates that *at least* two groups are different, serving as a trigger for more sophisticated analyses to identify exactly which groups differ.

21. How hypothesis testing is important in scientific research? A university professor claims that the standard deviation for students taking a research techniques test is less than 30. Ten tests are randomly selected, and the standard deviation is found to be 28.8. Test the professor's claim at the α= 0.01 level.
22. .How is textual data analyzed in research? Explain the basic theoretical frameworks used for qualitative analysis.
23. A researcher wants to measure student stress levels using a questionnaire. Calculate Cronbach's alpha and determine whether the scale is reliable.
24. Time Series Analysis and Multivariate Analysis

Time Series Analysis and **Multivariate Analysis** are advanced quantitative data analysis techniques used to identify complex patterns and relationships within research data.

Time Series Analysis

Time Series Analysis is the process of identifying and visualizing patterns in data collected over an extended period. It is fundamentally linked to **longitudinal research**, which observes the same subjects over time to track changes rather than taking a static "snapshot".

- **Core Concepts:** It focuses on **temporal order**, where researchers look for trends, cycles, or shifts that occur as a variable precedes another in time.
- **Interrupted Time-Series Designs:** In experimental research, two specific designs are used:
    - **Single-Group Interrupted Time-Series Design:** The researcher records multiple measures for a single group both before and after a treatment to see if the intervention caused a lasting change.
    - **Control-Group Interrupted Time-Series Design:** Two groups (not randomly assigned) are observed over time, but the treatment is only administered to one.
- **Visualization and Tools: Line graphs** are the primary tool for time series analysis, where time units are plotted on the horizontal axis (abscissa) and the target behavior is on the vertical axis (ordinate). Regularly published documents, such as government reports or technical logs, often serve as the secondary data source for longitudinal comparisons.

Multivariate Analysis

Multivariate Analysis involves the simultaneous analysis of **more than two variables** to identify complex patterns, interrelated outcomes, and causal models.

- **Role in CSE/IT Research:** In Computer Science and Information Systems research, outcomes like system productivity, code defect rates, and processing times are often related. Multivariate techniques allow researchers to examine these factors together rather than isolating them.
- **Key Techniques:**

    - **Multiple Regression:** Used to identify which independent variables are most strongly associated with an outcome.
    - **Logistic Regression:** A multivariate technique used specifically when the outcome variable is **binary** (e.g., "phishing" vs. "legitimate" or "success" vs. "failure").
    - **Multivariate Analysis of Variance (MANOVA):** Used when a study involves **multiple dependent measures** that must be tested against categorical independent variables.
    - **GEE Modeling and Mixed Models:** Advanced multivariate regression techniques used to compare changes over time between multiple groups.
- **Dimensionality and Complexity:** Sophisticated designs use these techniques to build elaborate models of causation, connecting multiple independent, intervening, and dependent variables into a single theoretical framework.

Summary Table: Comparison of Techniques

Feature	Time Series Analysis	Multivariate Analysis
Primary Focus	Tracking changes/trends over time.	Analyzing relationships between \(3+\) variables.
Research Design	Longitudinal or Interrupted Time-Series.	Experimental or Survey-based causal models.
Key Variable	Time (Temporal order).	Multiple Dependent/Independent variables.
Primary Goal	To capture ongoing processes.	To provide a holistic account of a complex problem.
Visualization	Line graphs of baseline vs. treatment.	Causal models, matrices, and regression tables.

a. A group of 6 employees completed a training program, and their performance scores were recorded before and after the training. 8

Before Training: 72, 75, 78, 70, 74, 77 After Training: 76, 79, 80, 73, 78, 81

Test whether the training program led to a significant improvement in performance scores. Also, specify which Python function(s) can be used to perform this test.

OR

A cloud service provider compares disk I/O rates (MB/s) for three storage systems over 4 days.

Storage System	Day 1	Day 2	Day 3	Day 4
S1	120	125	123	122
S2	135	138	136	137
S3	128	130	127	129

Use a one-way ANOVA to test whether there are significant differences in mean disk I/O rates among the storage systems. Interpret the results and explain their significance for storage selection

OR

25. A researcher uses $\alpha = 0.01$ for a test. Explain the implications of setting such a low significance level on Type I and Type II errors. In a study, a new drug is believed to reduce blood pressure. The average reduction in blood pressure for the population is 8 mmHg. A sample of 25 patients shows an average reduction of 9.5 mmHg. State the null and alternative hypotheses?

a. A researcher wants to measure student stress levels using a questionnaire. The scale contains 4 items (questions), each rated from 1 (low stress) to 5 (high stress). The researcher collects responses from 5 students: 8

Student	Q1	Q2	Q3	Q4
S1	4	5	4	4
S2	3	4	3	3
S3	5	5	4	5
S4	4	4	4	4
S5	2	3	2	2

To check the reliability of the scale, the researcher wants to know whether these 4 items consistently measure the same concept (stress). Calculate Cronbach's alpha and interpret whether the scale is reliable.